



Department of Computer Science and Engineering

Academic Year 2021 - 2022 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. Computer Science and Engineering

Course Code & Title: CS8601 & Mobile Computing

Name of the Faculty member (s): Ms.P.Jothi Thilaga, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Wireless Application Environment
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO12
- **Activity Chosen:** Flipped Classroom
- **Justification:**

Wireless Application Environment (WAE) is an important concept in mobile communications. By conducting the flipped classroom activity, the students initially learn the topic individually through the materials provided to them. Then the WAE topic is discussed in the class, which helps the students to get a comprehensive knowledge on this topic. Students studied individually about the WAE and share their ideas with their classmates, which also improves creative thinking and oral communication skills.

- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 - Learning materials such as video and PPT about WAE were sent to the student's mail and Canvas, one week before conducting this activity.
 - Instruct the students to watch the video individually at their home and take the notes according to their understanding level.
 - On the day of the activity, small teams were formed by their own interest.
 - Each student in a group discusses among the peer members in the concept of WAE based on their self-learning and write the points for Presentation within 10 minutes.
 - Once they prepared the content for presentation with their peer members, each group present about the different problems solved using backtracking algorithms for maximum of 5 minutes.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO10	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4
	3	2	2	2
Justification for correlation	Able to apply theory and principles of computer science and engineering to understand the wireless application environment	Able to use computer engineering principles to study the wireless application environment	Identify the concept of WAE to solve complex engineering problem	Choose appropriate feature of WAE to conduct investigations on complex problems
	PO10	PSO1	PSO2	PSO3
	2	1	1	1
	Able to explain about WAE and demonstrate its application in mobile communications	Able to practice different wireless networking features to solve complex problems.	Able to use wireless networking technologies to develop reliable IT solutions	Able to determine appropriate technologies to solve real world problems in Industry.

- Images / Screenshot of the practice:



Figure 1: Flipped Classroom Activity

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The students were actively participated.
 - Students were able to explored their knowledge on explaining the WAE and various features and applications of WAE.
 - This activity makes all the students to gain well knowledge on Wireless Application Environment.

- ❖ ***Benefit of the practice:***

- From this activity, the students can get more clarity in the particular topic by discussing and sharing their views with the other students in the class.
 - Learning at home automatically becomes student-centric rather than teacher-centric.

- ❖ ***Challenges faced in implementation:***

- Due to time constraints, only few group of students were made to share the concepts by a random call.
 - Some team members are hesitated and lack to share the concepts that they learnt and discussed.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/41_IT6801_FlippedClassroom.pdf
 - ❖ <https://www.ritrjpm.ac.in/images/computer-science/CS8603-FlippedClassroom.pdf>

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